Definition. *Let Γ a graph and use V to denote the set of its vertices. For a number $n \in \mathbb{N}$ and $W \subseteq V$, define the* configuration space with sinks *as*

$$F_n^{sink}(\Gamma) = \{(x_1, \dots, x_n) \in \Gamma^n \mid \text{ for } i \neq j,\ x_i \neq x_j \text{ or } x_i = x_j \in W\},$$

where the vertices in W are called the sink vertices.

{const}

Construction 1 (Cube complex $K_n(\Gamma)$)**.**

Let E and V be respectively the set of edges and the set of essential vertices of Γ, and give Γ the path metric such that each edge has length 1. We define each edge e in Γ to have two distinct half edges*, each of them pointing towards each vertex respectively. For each half edge h that comes from an edge that's not adyacent to any sink, we call $\partial_1 h$ the vertex incident to h and $\partial_0 h$ the edge given by the half edge. If e is an edge incident to one sink vertex, it only has one half edge associated to it, and then $\partial_0 h$ is the sink vertex next to it and $\partial_1 h$ is the non sink vertex adyacent to it. If e is an edge next to two sink vertices, then pick an orientation on e and call $\partial_0 h$ the starting vertex of e and $\partial_1 h$ the ending vertex of e.*

We define an abstract poset $P_n(\Gamma) = (P_n^{(0)}(\Gamma), P_n^{(1)}(\Gamma), \dots, P_n^{(k)}(\Gamma), \dots)$ where $P_n^{(k)}(\Gamma)$ is the set of k-faces of $P_n(\Gamma)$ defined by pairs (f, S) such that:

(a) $f : E \sqcup V \to \mathbb{N}$ *is a function.*

(b) $S = \{h_1, \dots, h_k\}$ *is a set consisting of exactly k half edges of Γ.*

(c) $\partial_1 h_i \in V$ *for* $i \in \{1, \dots, k\}$ *and if* $i \neq j$*, either* $\partial_1 h_i \neq \partial_1 h_j$ *or* $\partial_1 h_i = \partial_1 h_j \in W$*.*

(d) If e is adyacent to one or two sinks, $f(e) = 0$.

(e) $f(v) \in \{0, 1\}$ *for each* $v \in V \setminus W$ *and if* $\partial_1 h_i \in V \setminus W$*,* $f(v(h_i)) = 0$*.*

(f) $\sum_{a \in V \sqcup E} f(a) = n - k$*.*

Then, the 0-faces correspond to the configurations of particles among the graph such that there is no particles on edges incident to one or two sink vertices. The set S on a k- face, represent the independent movements of k particles on the graph. If h_i comes from an edge non adyacent to a sink, then it represent a movement from the edge to $\partial_1 h_i$. If h_i comes from an edges adyacent to one or two sinks, it represents a movement form the sink vertex to its adjacent vertex ∂h_i.

Given (f_1, S) and $(f_2, S \cup \{h\})$ with $s \notin S$, we say $(f_1, S) \prec (f_2, S \cup \{h\})$ if one of these cases holds:

(i)
$$f_1(s) = \begin{cases} f_2(a) + 1 & \text{if } a = \partial_0 h \\ f_2(a) & \text{otherwise} \end{cases}$$

(ii)
$$f_1(s) = \begin{cases} f_2(a) + 1 & \text{if } a = \partial_1 h \\ f_2(a) & \text{otherwise} \end{cases}$$

This is a partial order on the set of faces of $P_n((\Gamma))$ and we take it as the relation of being a face in $P_n((\Gamma))$. Observe that $P_n((\Gamma))$ is the face poset of a cube complex which we denote as $K_n((\Gamma))$.

Observation. *The items 2, 3 and 5 guarantee independence of the movements. This is:*

- *No two particles move along the same half edge (2).*
- *No two particles move toward the same non-sink vertex (3).*

- *No particle moves towar an occupied non-sink vertex (5).*

Given a vertex (x, φ) of $K_n(\Gamma)$, let $\mathcal{S}_\varphi$ the colllection of subsets S that satisfy the following conditions:

1. For each $h_i \in S$, $\partial_1 h_i \in V$.
2. $\partial_1 h_i \in V$ for $i \in \{1, \dots, k\}$ and if $i \neq j$, either $\partial_1 h_i \neq \partial_1 h_j$ or $\partial_1 h_i = \partial_1 h_j \in W$.
3. If $h \in S$, $\varphi(\partial_0 h) + \varphi(\partial_1 h) \geq 1$.
4. If $h \in S$ and $-h \in S$, $\varphi(\partial_0 h) + \varphi(\partial_1 h) + \varphi(\partial_1(-h)) \geq 2$.

{lemma}

Lemma 1. *The subsets $S \in \mathcal{S}$ are in one to one correspondence with the cells that contain x.*

Proof. A cell containing x is, given by a subset S and a function $\psi : E \sqcup V \to \mathbb{N}$ that fulfills the conditions (a) to (f) stated on Construction 1. The conditions (a) and (b) are 1 and 2 respectively. Observe that, for any (ψ, S), there exist one 1-cell (f, h) with $(\psi, \{\emptyset\}) \prec (f, h) \prec (\psi, S)$ for each $h \in S$. To prove 3, we take $(f, \{h\})$ one of these 1-cells. Then either $\varphi(\partial_0 h) = f(\partial_0 h) + 1$, or $\varphi(\partial_1 h) = f(\partial_1 h) + 1$. Hence $\varphi(\partial_0 h) \geq 1$ or $\varphi(\partial_1 h) \geq 1$, and the result follows for every $h \in S$.

To prove 4, we suppose $\mid S \mid \geq 2$ and take $h_1 \in S$ such that $h_2 = -h_1 \in S$ too. Observe that then, $\partial_0 h_1 = \partial_0 h_2$. It follows that there is a 2-cell $(f, \{h_1, h_2\})$ and also 1-cells (f_1, h_1) and (f_2, h_2) such that $(\varphi, \{\emptyset\}) \prec (f_1, \{h_1\}) \prec (f, \{h_1, h_2\})$ and $(\varphi, \{\emptyset\}) \prec (f_2, \{h_2\}) \prec (f, \{h_1, h_2\})$. We will prove that $\varphi(\partial_0 h_1) + \varphi(\partial_1 h_1) + \varphi(\partial_1 h_2) \geq 2$ in each of the following cases:

Case 1: When $\varphi(\partial_0 h_1) = 0$. Then by 3, $\varphi(\partial_0 h_1) \geq 1$ and $\varphi(\partial_0 h_1) \geq 1$. The claim follows.

Case 2: If $\varphi(\partial_1 h_1) = 0$ and $\varphi(\partial_1 h_2) \geq 1$. Using 3 on h_1 we obtain $\varphi(\partial_0 h_1) \geq 1$. Therefore $\varphi(\partial_0 h_1) + \varphi(\partial_1 h_1) + \varphi(\partial_1 h_2) \geq 1 + 0 + 1 = 2$.

Case 3: If $\varphi(\partial_1 h_2) = 0$ and $\varphi(\partial_1 h_1) \geq 1$. Similarly, we use 3 on h_2 to conclude $\varphi(\partial_0 h_1) = \varphi(\partial_0 h_2) \geq 1$. Thus $\varphi(\partial_0 h_1) + \varphi(\partial_1 h_1) + \varphi(\partial_1 h_2) \geq 1 + 0 + 1 = 2$.

Case 4: When $\varphi(\partial_1 h_1) = 0$ and $\varphi(\partial_1 h_2) = 0$. Since $(\varphi, \{\emptyset\}) \prec (f_1, \{h_1\})$, either $\varphi(\partial_0 h_1) = f_1(\partial_0 h_1) + 1$ and $\varphi(a) = f_1(a)$ for any other a or $\varphi(\partial_1 h_1) = f_1(\partial_1 h_1) + 1$ and $\varphi(a) = f_1(a)$ for any other a, but the latter would contradict our assumption. Hence $\varphi(\partial_0 h_1) = f_1(\partial_0 h_1) + 1$ and $\varphi(a) = f_1(a)$ for any other a. Particularly $\varphi(\partial_1 h_2) = f_1(\partial_1 h_2)$.

Hence $\varphi(\partial_0 h_2) = \varphi(\partial_0 h_1) = f_1(\partial_0 h_1) + 1 \geq 1$.

Now because of $(f_1, \{h_1\}) \prec (f, \{h_1, h_2\})$, $f_1(\partial_1 h_2) = f(\partial_1 h_2) + 1$ and $f_1(a) = f(a)$ for any other a, or $f_1(\partial_0 h_2) = f(\partial_0 h_2) + 1$ and $f_1(a) = f(a)$ for any other a, but since the first one would imply $\varphi(\partial_1 h_2) = f_1(\partial_1 h_2) = f(\partial_1 h_2) + 1 > 1$, this is a contradiction and $f_1(\partial_0 h_2) = f(\partial_0 h_2) + 1$. This implies $\varphi(\partial_0 h_1) = f_1(\partial_0 h_1) + 1 = f_1(\partial_0 h_2) + 1 = (f(\partial_0 h_2) + 1) + 1 \geq 2$.

Now let S such that it fullfills 1 to 5. We construct the function $g : S \to \{-1, 0\}$ that assigns this label to each of the half edges as follows. If $\varphi(\partial_1 h) \leq 1$, $g(h) = (\varphi(\partial_1 h) - 1)$. This is, -1 if $\varphi(\partial_1 h) = 0$ and 0 otherwise. When h is such that $\varphi(\partial_0 h) > 1$ and $\varphi(\partial_1 h) > 1$, it randomly assigns 0 or -1 to h.

Then the set S corresponds to the cell (f, S) where function f is given by:

$$f(a) = \begin{cases} \varphi(a) & \text{if } a \in V \setminus \{\partial_0 h \mid h \in S\} \cup \{\partial_1 h \mid h \in S\} \\ 0 & \text{if } a \in \{\partial_1 h \mid h \in S\} \setminus (W \cap \{\partial_1 h \mid h \in S\}) \\ \varphi(a) + \sum_{\{h \in S \mid \partial_0 h = a\}} (\varphi(\partial_1 h) - 1) & \text{if } a \in E \\ \varphi(a) + \sum_{\{h \in S \mid \partial_i h = a, i = 0, 1\}} g(h) & \text{if } a \in (\{\partial_1 h \mid h \in S\} \cup \{\partial_0 h \mid h \in S\}) \cap W \end{cases}$$

□

Proposition 1. *The link of each of the vertices of the cube complex K is a flag complex.*

Proof. Let $x \in K$ a vertex of the cube complex. This is, a configuration where all particles in the interior of each edge are equally spaced and there is no particle on an edge adjacent to one or two sinks. Formally, this is $x = (\varphi, \emptyset)$ where φ is a function $\varphi : E \cup B \to \mathbb{N}$ such that:

(a) $\varphi(v) \in \{0, 1\}$ for each $v \in V \setminus W$

(b) If an edge e is incident to one or two sink vertices, $\varphi(e) = 0$.

(c) $\sum_{a \in V \sqcup E} \varphi(a) = n$

We'll describe the link of x. Start by the cells that contain x, this is, the faces F of the cube $K_n(\Gamma)$ that have x as one of its face. To describe them, we can instead use Lemma 1 to identify them with the element of the $\mathcal{S}$.

The subposet of cells in $K_n(\Gamma)$ which contain x coincides with the poset P_φ of sets in $\mathcal{S}_\varphi$ ordered by inclusion. This poset has the property that if $S \in P_\varphi$ and $\emptyset \neq R \subset S$ then $R \in P_\varphi$. This implies that P_φ is the face poset of a simplicial complex K_φ, whose k-simplicies correspond to the elements $S \in P_\varphi$ with cardinality $k + 1$. The poset P_φ is isomorphic to the face poset of the link, and then this simplicial complex is isomorphic to the link.

A set $S \in P_\varphi$ satisfies the conditions 1-4, and these have the property that a set S of half edges satisfy them if and only if each subset of S with cardinality 2 does. Then this is a flag complex $\square$

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